# Alexandra Schuller, CPI

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## SKILLS AND QUALIFICATIONS

**Languages** Bilingual in French and English, with native-level proficiency in both.
**Modelling and Simulation** | Abaqus • DIANA • SAP2000 • HEC-RAS • AutoCAD • CSI Bridge • ETABS • SAFE • Revit • SUMO • WoodWorks
**Programming** | MATLAB • Python • VBA (Excel) • Octave • GitHub • Claude Code • WordPress
**Standards** | CSA S6 • CSA S7:23 • CSA A23.3 • CSA S16 • CSA O86 • NBCC 2020
**Project Management** | MS Project • Primavera P6 • Procore • Aconex • Bluebeam • Stack • Asana • Notion
**Other** | Accpac/Sage • Concur • Windows • Linux • MacOS
**Licences & Certifications** | Driver's licence • ASP Construction Health and Safety (2022) • ASP Lockout • WHMIS for laboratory personnel (Concordia)

## EDUCATION

**Bachelor of Engineering, Civil Engineering (Infrastructure Specialisation), with Distinction** **2022-2026**
**Co-operative Education Program (Co-op)**
Concordia University, Montreal, QC
Member of the Institute for Co-operative Education

**University Prerequisite Program for Adults, Continuing Education** **2021-2022**
Dawson College, Semester Honours

**Bachelor of Arts, Classical Studies, Specialization in Archaeology** **2009-2013**
McGill University, Montreal, QC

**DEC in Arts and Sciences** **2007-2009**
Marianopolis College, Montreal, QC

## DESIGN PROJECTS

**Rock Slope Stability Analysis, Northern Quebec** **January - April 2026**
Course project, Advanced Geotechnical Engineering (CIVI 437), Concordia University
Stability analysis from triaxial test data using the Hoek-Brown failure criterion, converted to equivalent Mohr-Coulomb parameters (Excel, MATLAB), and finite element modelling in Abaqus; assessed the effect of GSI (100 and 85) on cohesion, friction angle and stability.

**CapDécarie, Civil Engineering Capstone Project** **September 2025 - April 2026**
Took the initiative to propose a project outside the options offered, then recruited and rallied a team of five around the concept: covering a section of the Décarie Expressway trench, between Snowdon and Queen Mary, with a 5,670 $m^2$ pedestrian park, and replacing the aging Queen Mary overpass.

- Responsible for the design of the foundations and beams of the cap structure (a deck of about 180 m), the drainage design and the traffic simulation for construction staging.
- Redesigned the Queen Mary overpass and validated structural behaviour with finite element analysis (CSI Bridge, SAP2000); steel and reinforced concrete design, load analysis and seismic analysis.
- Also covered embodied carbon, cost estimating and construction planning.

**Captain, Concordia Concrete Canoe Team** **September 2023 - May 2026**
Revived the team after seven years of inactivity and led twelve people in designing, building and transporting a 6-metre concrete canoe to the national competition in Quebec City.

- Concrete mix design, laboratory testing and structural analysis of the canoe.

**10-Storey Reinforced Concrete Office Building, Quebec City** **September - December 2025**
Course project, Design of Reinforced Concrete Structures (CIVI 453), Concordia University
Complete structural design of a reinforced concrete office building (flat slabs, columns and ductile shear walls) to NBCC 2020 and CSA A23.3: wind and seismic loads, modelling in ETABS and SAFE.

## WORK EXPERIENCE

**Intern, Direction générale des barrages** **April - August 2025**
**Major Projects Directorate**
**Ministry of the Environment, the Fight Against Climate Change, Wildlife and Parks (Quebec)**
Helped implement a new centralized MS Project planning system for about 1,000 dams in Quebec by consolidating project schedules and milestones with Python and AI tools. Site visits and introduction to major work on new and existing dams.

**Intern, Renouveau Lafontaine - Louis-Hippolyte-La Fontaine Tunnel** **January - April 2025**
**Pomerleau**
Tracked the progress of heating cable repairs in the main and access tunnels; built a Python tree of the tunnel's electrical systems that automatically generates their power supply diagrams.

**Intern, Local Energy Efficiency Partnerships (LEEP)** **September - December 2023**
**Natural Resources Canada**
**CanmetENERGY, Ottawa**
Designed a VBA data analysis tool (costs, GHG, energy) for converting homes to heat pumps, which uncovered an error in published government data and led to a revision.

**Program Assistant - The McConnell Foundation** **January 2015 - August 2021**
**Montreal, QC**
Supported the Cities for People program, which funds projects that improve urban life across Canada.

**Intern & Research Assistant - Université de Montréal** **Summers 2011 and 2012**
Archaeological excavations at *Argilos*
Surveyed the site, laid out trenches and documented the excavations following archaeological protocols.

## AWARDS & DISTINCTIONS

**Graduated with Distinction - Bachelor of Engineering, Civil, Concordia University** **June 2026**

**Pierre Sauvé, Eng. Scholarship for the Advancement of Women in Engineering** **March 2023**
Fondation de l'Ordre des ingénieurs du Québec (FOIQ)

**Valedictorian - Centennial Regional High School (CRHS)** **June 2007**

## PROFESSIONAL ASSOCIATIONS

**Candidate to the Engineering Profession (CPI) - Ordre des ingénieurs du Québec** **September 2026**

**Board Member** **December 2022 - May 2023**
The Concordia Greenhouse, Montreal, Quebec
Member of the infrastructure committee overseeing options to upgrade the greenhouse's existing components.

## OTHER INTERESTS

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| **Sports** | Hockey, softball and tennis; Triathlon Montréal Esprit Sprint (2020) |
| **Other Passions** | Greenhouse design, archaeology, community gardens, music (200+ public performances) |

## REFERENCES

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|---|---|
| **Dr. Archie Huang** | Assistant Professor, Building, Civil, and Environmental Engineering • archie.huang@concordia.ca |
| **Dr. Adel Hanna** | Professor, Building, Civil, and Environmental Engineering • am.hanna@concordia.ca |